

Halogens in hydrothermal sphalerite record origin of ore-forming fluids

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ABSTRACT

The halogens Cl and Br are sensitive indicators for the origin of ore-forming fluids. Here, we use a combination of microchemical and microscopic methods to show that measurable concentrations of these elements commonly occur as atomic-scale substitutions in hydrothermal sphalerite. Furthermore, the Cl/Br ratios of halogen-rich sphalerites are indistinguishable from those of the corresponding ore-forming fluids. Thus, they record fluid compositions, which in turn record fluid origin. Given the abundance of sphalerite in hydrothermal base-metal deposits, as well as the relative ease of conducting *in situ* microchemical analyses, the halogen signature of sphalerite has the potential to become a sensitive proxy to distinguish between different ore-forming environments.

INTRODUCTION

Due to their ability to form stable complexes with many metals in aqueous solutions, halogens play a key role in mass-transfer processes on Earth (Harlov and Aranovich, 2018). Not only are they heavily involved in the formation of most hydrothermal ore deposits (Pirajno, 2018), but they also critically affect the nature and rate of various magmatic and metamorphic processes (Dolejš and Zajacz, 2018; Hammerli and Rubenach, 2018).

Because they generally behave conservatively during fluid-rock interaction and show distinctive signatures for different crustal reservoirs, Cl and Br have seen extensive use as tracers for the origin of ore-forming hydrothermal fluids (Burisch et al., 2016). While previous studies primarily relied on the analysis of either fluid inclusions or scarce silicate minerals such as scapolite to constrain fluid compositions

(e.g., Kesler et al., 1995; Hammerli et al., 2013), this article is the first to demonstrate the *in situ* measurement of Cl and Br in sphalerite.

Sphalerite is common in many types of hydrothermal base-metal deposits. It is well known for its diverse trace element chemistry (Cook et al., 2009; Belissont et al., 2014), recording critical information about the physicochemical conditions of ore formation (Frenzel et al., 2016). Although high concentrations of Cl in sphalerite have previously been reported (Taylor and Radtke, 1969; Barrie et al., 2009), there is a lack of systematic data on the general prevalence of Cl and other halogens in natural samples, as well as the likely implications of this phenomenon.

Because they are not part of the classic suite of chalcophile elements, the halogens are not commonly included in the set of elements analyzed in sulfide minerals. This is despite the fact

that (1) the Cl⁻ ion is virtually the same size as S²⁻ (Shannon, 1976) and would therefore be expected to readily substitute into sulfide minerals (Blundy and Wood, 2003); and (2) Cl/S ratios in hydrothermal fluids are generally high (Yardley, 2005). Of the other halogens, particularly Br⁻ would be expected to substitute for S²⁻ for similar reasons. These considerations provided the major motivation for this work. As our data show, halogen substitution into hydrothermal sphalerite may be a common phenomenon, particularly in low-temperature deposits.

MATERIALS AND METHODS

To study the occurrence of Cl and Br in natural sphalerite, we selected 12 samples from a diverse range of ore deposits (Table 1). Chlorine concentrations were first measured in all samples using electron probe micro-analysis (EPMA), because this technique offers the best detection limits for Cl. EPMA measurements were supplemented by laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS) to independently verify the presence of Cl and determine the abundances of Br and other relevant trace elements (Ga, Ge, In, Co, Tl, Na, etc.). Samples with high Cl concentrations were then selected for detailed mapping by EPMA (micrometer scale) and scanning transmission electron microscopy (STEM; nanometer scale) coupled with energy-disper-

¹Supplemental Material. Appendices S1 (detailed methods and summary of analytical results), S2 (full EPMA dataset), S3 (full LA-ICP-MS dataset), and S4 (fluid database used for Figure 3). Please visit <https://doi.org/10.1130/G47087.1> to access the supplemental material, and contact editing@geosociety.org with any questions.

TABLE 1. CL, BR, AND NA CONCENTRATIONS IN SPHALERITE FROM DIFFERENT DEPOSITS (MEDIAN VALUES)

Sample ID	Deposit	Deposit type*	Location		Sphalerite generation	Cl (μg/g)	Br (μg/g)	Na (μg/g)
			Latitude	Longitude				
RX7319	Olympic Dam, Australia	IOCG	30°26'S	136°52'E	I	B.D.L.	N.D.	B.D.L.
BBH19	Baita Bihor, Romania	Skarn	46°28'N	22°34'E	I	B.D.L.	B.D.L.	B.D.L.
BS7b	Baisoara, Romania	Skarn	45°51'N	24°58'E	I	1200	21	64
TM06.2	Tres Marias, Mexico	MVT	29°05'N	103°45'W	I	230	3.1	17
					II	46	2.5	N.D.
Toy-1	Toyoha, Japan	Vein	42°58'N	141°02'E	I	B.D.L.	N.D.	N.D.
S4	Efemçukuru, Turkey	Vein	38°17'N	26°58'E	I	B.D.L.	B.D.L.	B.D.L.
Zn99.2	Zinkgruvan, Sweden	SHMS	58°48'N	15°06'E	I	B.D.L.	N.D.	B.D.L.
TSU	Tsumeb, Namibia	MVT	19°13'S	17°43'E	I	B.D.L.	N.D.	B.D.L.
FG	Freiberg, Germany	Vein	50°55'N	13°21'E	I	B.D.L.	N.D.	B.D.L.
Li-HS-86	Lisheen, Ireland	MVT	52°44'N	07°40'W	I	180	B.D.L.	B.D.L.
					II	4900	23	310
65	Plaka, Greece	HTCR	37°45'N	24°01'E	I	B.D.L.	B.D.L.	B.D.L.
MP	Mavres-Petres, Greece	HTCR	40°31'N	23°47'E	I	B.D.L.	B.D.L.	B.D.L.

Note: Cl values are from electron probe micro-analysis (EPMA), and Br and Na come from laser ablation–inductively coupled plasma–mass spectrometry (LA-ICP-MS) analyses. Full details of the data are provided in Appendices S1, S2, and S3 (see text footnote 1). B.D.L.—below detection limit (~30 μg/g for Cl; 0.3–20 μg/g for Br; and 0.1–8 μg/g for Na, depending on specific measurement conditions); N.D.—no data.

*Deposit type abbreviations: HTCR—high-temperature carbonate replacement; IOCG—iron-oxide-copper-gold; MVT—Mississippi Valley type; SHMS—sediment-hosted massive sulfide; Vein—epithermal vein.

sive X-ray spectroscopy (EDX) to study the spatial distribution of Cl at different scales and determine its mode of occurrence (substitution or inclusions). Full details of all analytical methods are given in Appendix S1 in the Supplemental Material¹.

RESULTS

EPMA measurements showed that appreciable Cl concentrations occur in three out of the 12 investigated samples, ranging from 46 to 4900 μg/g on average (Table 1). In most samples, however, Cl concentrations were below the detection limit (~30 μg/g). Where Cl concentrations exceeded ~100 μg/g, they were also measurable by LA-ICP-MS and were accompanied by measurable Br and Na concentrations. Laser ablation traces of Cl, Br, and Na were generally smooth (Fig. S1 in the Supplemental Material), indicating homogeneous distribution of these elements within the sphalerite matrix at the micrometer scale.

Exploratory analysis of the micro-chemical data set showed that the halogen-rich sphalerites are characterized by significantly higher Ti and Ge concentrations and lower Mn concentrations compared to the halogen-poor ones (Appendix S1). Other trace elements (Cd, Co, Fe, Ga, In, etc.) do not show significant correlations with halogen content.

Due to their very high halogen content, the samples from Lisheen, Ireland (Li-HS-86) and Baisoara, Romania (BS7b) were selected for further investigation by EPMA mapping and STEM-EDX imaging. Sample Li-HS-86 is dominated by fine-grained, color-banded colloform sphalerite, while sample BS7b features more coarsely crystalline black to yellowish-translucent sphalerite.

Figure 1 illustrates the results of EPMA mapping for sample BS7b. The corresponding figure for sample Li-HS-86 is included in Appendix S1 (Fig. S6). In general, the EPMA maps show that

(1) Cl distribution from the millimeter to micrometer scale is characterized by complex, in some cases oscillatory, growth zoning qualitatively like that shown by the other investigated elements (Cd, Cu, Pb, Fe, where measurable); and

(2) In the highest-resolution EPMA maps (step size: 1 μm), Cl is homogeneously distributed within zones of similar composition. There is no evidence for Cl-rich inclusions at this scale.

Further examination of Cl-rich areas by high-angle annular dark field (HAADF)–STEM imaging showed that both samples have complex nanostructures, featuring nanoscale porosity, solid inclusions, and sub-grain boundaries (Fig. 2; Fig. S7). In addition, STEM-EDX mapping and semiquantitative EDX analysis allowed for the examination of Cl distribution at the nanoscale. Here, there are marked differences between the two samples:

- Sample BS7b shows no evidence of Cl-rich inclusions. Instead, Cl appears to be homogeneously distributed throughout the material at the nanometer scale (1–5 nm). No Cl enrichment is associated with any of the observed porosity, sub-grain boundaries, or solid inclusions (Fig. 2).
- Sample Li-HS-86 contains a substantial amount of Pb-Cu-Cl-rich inclusions that are commonly associated with porosity. Nevertheless, an appreciable amount of Cl is still homogeneously distributed throughout the material at the scale of 1–5 nm (Fig. S6). A mass-balance calculation using LA-ICP-MS data for Cu, Pb, and Cl indicates that only 25% of the total Cl content in sample Li-HS-86 is hosted in Pb-Cu-Cl inclusions (cf. Appendix S1, section 2.6).

Overall, both samples appear to contain considerable amounts of Cl that is homogeneously distributed throughout the sphalerite at a scale of 1–5 nm. Unfortunately, the micrometer- and

nanometer-scale distributions of Br and Na could not be investigated in detail by STEM-EDX or EPMA due to their considerably lower concentration levels (rarely exceeding tens of micrograms per gram; cf. Table 1), as well as interference from other elements (e.g., Zn on Na). However, no Br- or Na-containing nano-inclusions were found in either of the two samples.

DISCUSSION

Below, we briefly discuss the implications of our observations for the incorporation mechanism(s) of Cl, Br, and Na in sphalerite. We also highlight potential geological applications for in situ measurements of Cl and Br.

Incorporation Mechanisms of Cl, Br, and Na

In principle, Cl, Br, and Na can occur in hydrothermal sphalerite in three different forms: (1) as atomic-scale substitutions for S and Zn, (2) as nano- or microscale inclusions of solid phases, and (3) in inclusions of hydrothermal fluids. The analytical data we collected provide constraints on the relative contributions of each of these three forms.

Available evidence suggests that most Cl is present as atomic-scale substitutions. The major arguments for this are the homogeneous distribution of most Cl at a scale of 1–5 nm, and the large excess of Cl compared to Na in the sphalerite.

A distance of 1–5 nm corresponds to only two to 10 unit cells of the sphalerite structure ($c = 0.54$ nm). A quick calculation shows that a $2 \times 2 \times 70$ nm volume of ZnS (four pixels in the EDX maps in Fig. 2D, times sample thickness) at an average Cl concentration of 0.3 wt% would contain only four Cl atoms. Thus, the homogeneous distribution of Cl across pixels of this size is good evidence for its presence as atomic-scale substitutions. Areas homogeneous at this scale exist on both samples BS7b and Li-HS-86. However, sample Li-HS-86 also contains a substantial amount of Cl in solid inclusions (~25%; cf. Appendix S1).

The nanostructural evidence for atomic-scale substitution of Cl is corroborated by the fact that the molar ratio of Cl to Na in all halogen-rich samples is ~10 (Table 1). This means that inclusions of the ore-forming fluids cannot be the major hosts of Cl in any of these samples, because such fluids should have a Cl/Na ratio of ~1 (Yardley, 2005).

For Br, smooth LA-ICP-MS traces (Fig. S1) suggest homogeneous distribution at the scale of ~1 μm. The additional absence of detectable Br-rich nano-inclusions during STEM-EDX also suggests its presence as atomic-scale substitutions. This is corroborated by the lack of fractionation between Cl and Br during partitioning from the ore-forming fluids into sphalerite (see below; Fig. 3), which implies that

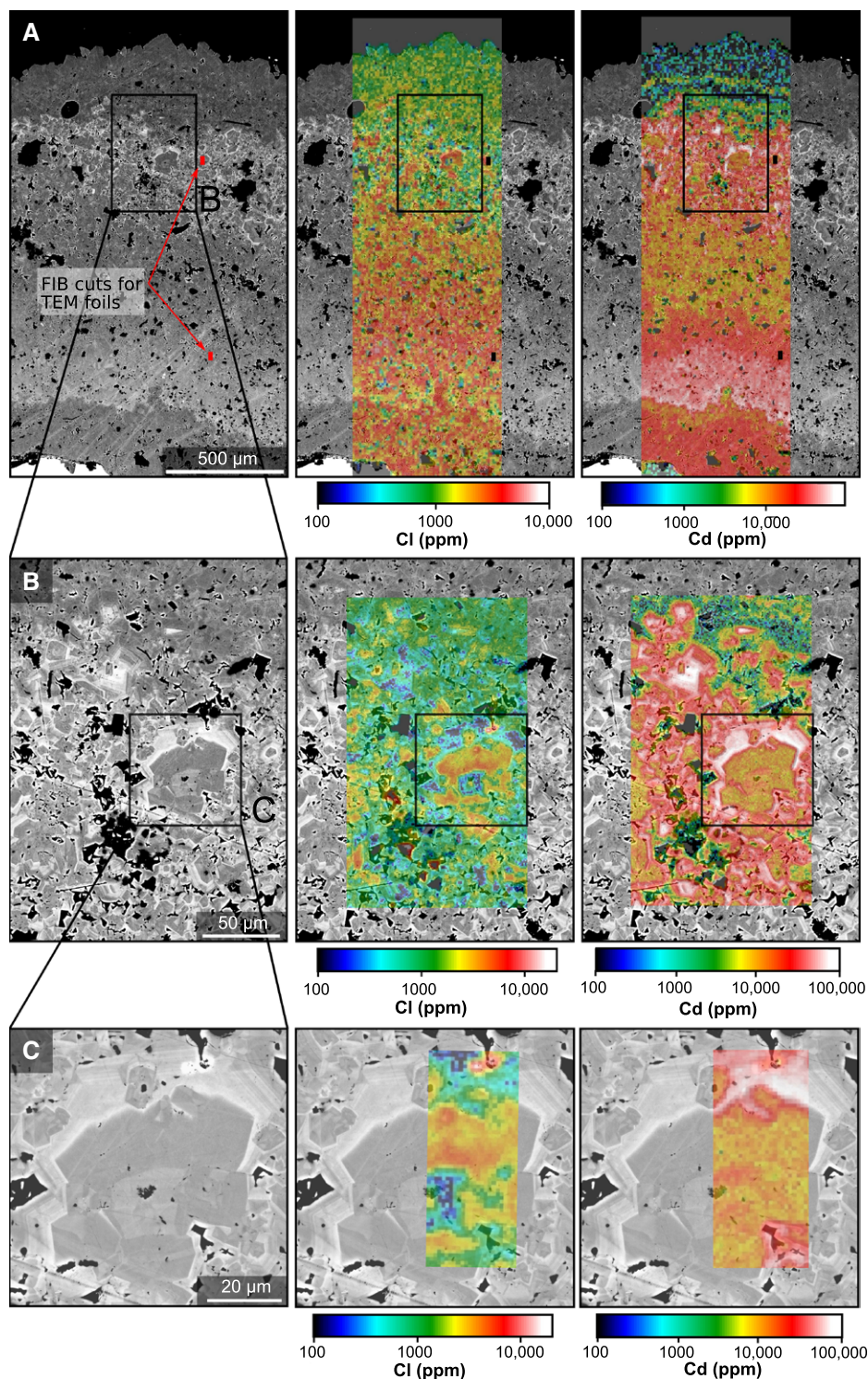


Figure 1. Electron probe micro-analysis maps at different scales for sphalerite sample BS7b (Baisoara, Romania) showing Cl and Cd distribution, superimposed on high-contrast back-scattered electron images of the polished sample surface. (A) Millimeter-scale distribution. (B) Intermediate-scale distribution. (C) Micron-scale distribution. FIB—focused ion beam; TEM—transmission electron microscopy.

the two elements are incorporated through the same mechanism.

Finally, Na is the element with the least well-constrained mode of occurrence. While it is also homogeneously distributed at the scale of $\sim 1 \mu\text{m}$, analytical difficulties preclude its

reliable detection by STEM-EDX in a ZnS matrix. This means a definitive statement about its mode of incorporation is currently not possible. It could be present in nanoscale fluid inclusions. This would constrain the proportion of Cl hosted by such inclusions to $<10\%$ of the total

Cl content, assuming the Cl/Na ratio in the fluid inclusions is $\sim 1 \text{ mol/mol}$ (Yardley, 2005). However, Na could also be present as atomic-scale substitutions for Zn.

A discussion of the implications that lattice-hosted Cl, Br, and potentially Na would have for the crystal chemistry of sphalerite is included in Appendix S1.

Potential Applications

Two major geological applications can be envisaged for measurements of Cl and Br concentrations in sphalerite. First, they may be useful as tracers of fluid origin. Second, it may be possible to use them as indicators for fluid salinity. Together with existing sphalerite geothermometers (Keith et al., 2014; Frenzel et al., 2016), this would enable the characterization of formation temperature, fluid salinity, and fluid origin for a sphalerite sample with only a small set of in situ microchemical analyses. Given the current need to combine time-consuming microthermometric measurements with at least one other technique (LA-ICP-MS and/or crush-leach analyses) to yield the same information (Kesler et al., 1995; Seo et al., 2011; Hammerli et al., 2013), this would constitute a substantial step forward.

Figure 3 shows that Cl/Br ratios in the investigated sphalerite samples are close to the ratios of the corresponding ore-forming fluids. This indicates that no major fractionation occurs between Cl and Br when partitioning from the fluid into the sphalerite, making them suitable as tracers of fluid composition and origin. Agreement between sphalerite and fluid composition is particularly good for samples Li-HS-86 and TM06.2, which show distinct Cl/Br ratios yet lie within the range expected for their corresponding ore-forming fluids (Irish-type and Mississippi Valley-type fluids). Only in sample BS7b is the Cl/Br ratio lower than expected for a skarn fluid. However, it still lies within the range covered by the fluid data (including outliers). The skarn data available for comparison derive from only two ore deposits. Therefore, the true range for these deposits is likely broader. Finally, the formation temperature of sample BS7b is very low for a skarn deposit (cf. Appendix S1), and it may therefore not be typical for this deposit class.

Another tentative application for Cl in sphalerite is its use as a proxy for fluid salinity. To first order, absolute concentrations of lattice-hosted Cl should be controlled by the respective concentrations in the ore-forming fluid (McIntire, 1963). Because Cl^- is the major anion in crustal fluids (Yardley, 2005), such relationships should be usable in a sphalerite salinometer. However, the fact that high Cl concentrations are present only in low-temperature sphalerites (cf. Appendix S1) suggests that some temperature control may also exist. A correction for such a

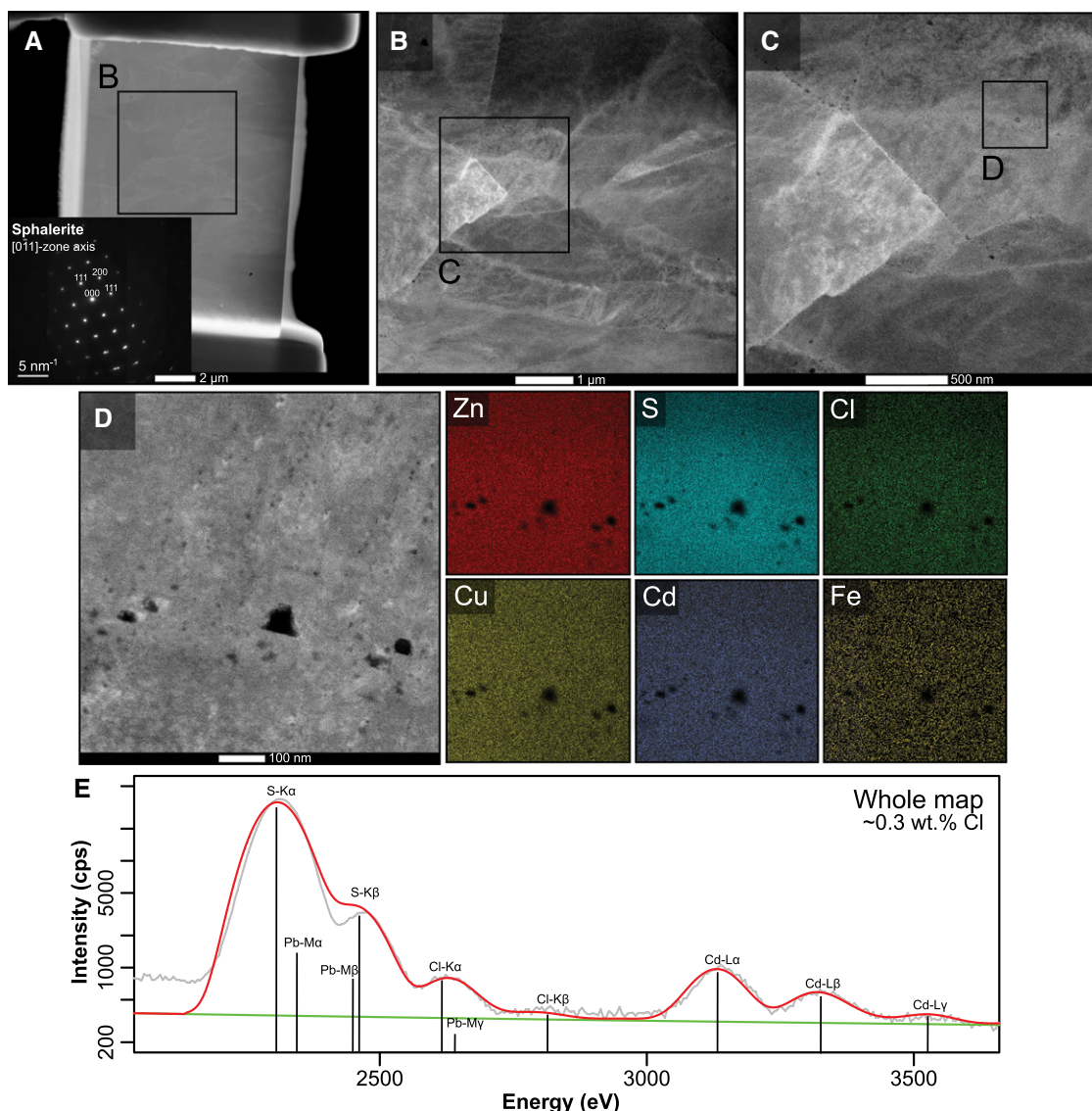


Figure 2. Summary of scanning transmission electron microscopy (STEM) and STEM-energy-dispersive X-ray spectroscopy (EDX) imaging in sphalerite sample BS7b (Baisoara, Romania). (A) Overview of sample foil with corresponding selected area electron diffraction (SAED) pattern in the lower left-hand corner. (B–D) Detailed high-angle annular dark field (HAADF)-STEM images of A as indicated, with EDX maps corresponding to D. (E) Summed EDX spectrum for the entire map area shown in D. Gray line is data; red line represents fitted spectrum; green line represents fitted background. Note the apparent nanostructure of the material (different domains, pores, brightness differences) in D, as well as the chemical homogeneity shown by EDX maps, particularly for Cl concentration.

dependence would need to be incorporated into any potential salinometer. This will require the collection of additional data on samples where the ore-forming fluids are well characterized by fluid inclusion microthermometry in terms of their temperature and salinity.

Overall, the use of element ratios such as Cl/Br shows the greatest potential for direct application in economic geology. Sphalerite salinometry may become feasible in the future as more analytical data become available, allowing for the correction of potential temperature effects on Cl incorporation. In both applications, care should be taken in the selection of samples to ensure that halogens are present in a form that records the desired signal. For Cl/Br ratios, both atomic-scale substitutions and nanoscale fluid inclusions should reliably record the signal of the ore-forming fluids. For salinometry, however, only lattice-hosted Cl would be relevant. In general, we would expect coarsely crystalline samples like sample BS7b to fulfill the conditions for

both applications. However, colloform samples like sample Li-HS-86 cannot be assumed to be suitable in all cases, because they may contain abundant solid nano-inclusions.

CONCLUSIONS

Atomic-scale substitution of Cl and Br for S in hydrothermal sphalerite commonly leads to measurable concentrations of the two elements, particularly in base-metal deposits formed at low temperatures (<200 °C). Furthermore, Cl and Br are incorporated without fractionation, such that the Cl/Br ratio of sphalerite records the signature of the ore-forming fluids, in turn allowing for the identification of the fluids' origin.

In the future, it may also be possible to infer fluid salinity from absolute Cl concentrations in sphalerite. In combination with existing sphalerite geothermometers, both applications would make the trace-element signature of sphalerite an invaluable indicator in the study of hydrothermal ore deposits. Its major advantages com-

pared to current methods used to determine the temperature, salinity, and halogen signature of ore-forming fluids are the relative ease of sample preparation and analysis compared to fluid inclusion studies, the generally much greater abundance of sphalerite than scapolite in many base-metal deposits, as well as the better control over the geological context and significance of the sample material compared to crush-leach analyses.

ACKNOWLEDGMENTS

We gratefully acknowledge support from the German Academic Exchange Service (DAAD) in funding the first author's stay in Adelaide, where most of this work was conducted. The geoscientific collections of the Technische Universität Bergakademie Freiberg (Germany) are thanked for providing samples TSU and FG, while Vedanta Zinc International (Johannesburg, South Africa) is thanked for providing sample Li-HS-86. Finally, we thank Stephen Kesler, Jung-Hun Seo, Jens Gutzmer, and an anonymous reviewer for their constructive comments that helped to significantly improve this manuscript.

Cl/Br ratios of hydrothermal fluids and sphalerite

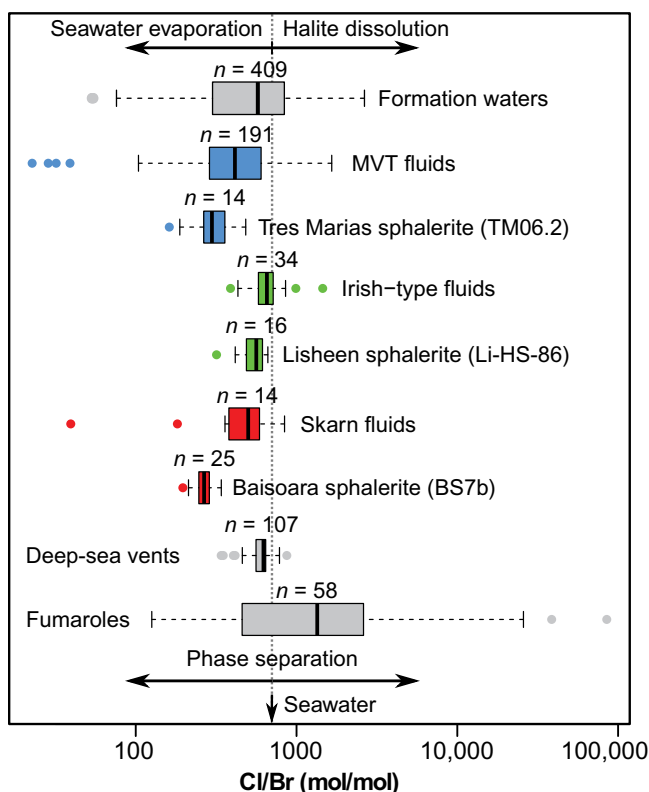


Figure 3. Boxplot comparing Cl/Br ratios in halogen-rich sphalerite samples from this study (see Table 1) with literature data on relevant ore-forming and crustal fluids (see Appendix S4 [see footnote 1] for the complete fluid data set including a reference list). Lower and upper limits of the boxes correspond to the 1st and 3rd quartiles of the data, respectively. Vertical line in each box represents the median value. Whiskers extend to the most extreme data points lying within 1.5x the interquartile range of each box. Dots represent data points lying outside of this range. The colors red, green, and blue are intended as a visual aid, showing each sphalerite sample and the corresponding group of fluid analyses in the same color. Fluid populations in gray are shown for a more general comparison. Lisheen is located in Ireland; Baisoara is located in Romania; Tres Marias is located in Mexico. MVT—Mississippi Valley-type.

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